

David Tran

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Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash/Shell

Cloud & Infrastructure: AWS (S3, DynamoDB, EC2, Lambda, CloudFront, API Gateway, IAM), Infrastructure as Code (CloudFormation, CDK), Observability / Monitoring (CloudWatch), Linux, CI/CD, Docker

Technologies: Distributed Systems, RESTful APIs, Microservices, Authorization (RBAC), Git

Experience

Software Engineer (Independent Project), Automated Trading Platform – San Diego, CA July 2025 – Feb 2026

- Built and operated a personal automated trading system, focusing on reliability, risk controls, and fault tolerant execution for event-driven trade execution
- Developed and deployed a Python service on EC2 to ingest TradingView webhooks, validate JSON payloads, and execute trades via exchange APIs
- Prevented duplicate and conflicting trades by enforcing a single-position constraint through live exchange state checks before order placement
- Improved robustness against external API failures by adding retry logic, input validation, structured logging, and alerts; validated behavior across 100+ webhook events and 100+ paper trades

Software Development Engineer, Amazon – Seattle, WA May 2021 – Jun 2024

- Cut edge device onboarding times from 3–6 hours to 1 hour and reduced errors by 60% designing and building backend APIs and orchestration workflows for a self-service device provisioning platform, automating OS install, network config, and software deployment
- Reduced hardware sourcing lead times from 10–16 weeks to 2 weeks ahead of Prime Day by validating an alternative device on-site and extending provisioning workflows to support multi-vendor hardware, enabling global deployment of 250+ devices across Amazon Operations
- Reduced CI/CD build and test cycles by 40% through refactoring services, tests, pipelines, and CloudFormation-backed environments; improved reliability across Amazon edge devices by adding EC2/VirtualBox simulations and CloudWatch dashboards/alerts with runbooks
- Scoped and implemented key components of AuthZ rollout for the provisioning platform, enabling per-client RBAC and least-privilege access; guided an intern through implementation and executed backfill migration and automated permission provisioning for new tenants
- Reduced provisioning-related tickets for warehouse automation teams by scripting automated validation and redeploy of AWS IoT Greengrass Docker components and documenting common failures in runbooks

Junior Software Engineer, Inseego – San Diego, CA Oct 2019 – Mar 2020

- Transitioned a vendor-owned Azure workflow into Inseego's environment by reverse-engineering the existing implementation and rebuilding components where no direct migration path existed
- Created test files and wrote ad-hoc SQL queries to validate releases and support QA during development

STEM Mentor, Dreams for Schools – Irvine, CA Sep 2018 – Dec 2018

- Led a student team to 1st place out of 22 teams in a showcase competition at UC Irvine
- Taught students to build STEM-focused Android applications over a 9-week program

Education

University of California, Irvine – Bachelor of Science in Computer Science

Mar 2020